Supplement — Where the Upper Bound Comes From

OL7014 Integer Programming • Instructor: Prof. H. Park • companion to Week 7, §§4–5 • Oct 12, 2026

1. One relaxation per node — that is the whole construction

week 7: node k owns the piece $X_k = P_k \cap \mathbb{Z}^n_+$, and LP_k relaxes it. *that LP value is the node's bound*

$$z^k \;=\; \max\{\, c^\top x : x \in X_k \,\} \;\le\; z_{\mathrm{LPR}}(k) \;=\; \max\{\, c^\top x : x \in P_k \,\}$$

$$P_k \;=\; P_0 \cap \{\, x_j = v \;:\; (j,v) \text{ fixed on the path down to } k \,\}$$

no second model, no multipliers: **constructing the bound = solving one LP**

in the code the LP is built *once*, with the binaries relaxed to $0 \le x_i \le 1$; branching never rebuilds it — it only moves two numbers:

$$\text{fix } x_j = v \;\text{ on the path} \quad\Longrightarrow\quad \ell_j = u_j = v, \qquad \text{re-solve}$$

each child differs from its parent in one bound, so the simplex restarts warm — which is why thousands of nodes are affordable

KEY a bound is a *promise about a region*: $z_{\mathrm{LPR}}(k)$ says "nothing in X_k can beat this", and it is cheap because the region is described by bounds alone.

2. The number on the screen — the best promise still outstanding

at any moment the tree holds **closed** nodes (pruned or branched) and **open** nodes (created, not yet solved)

the global upper bound (상한) is the best promise over the open nodes — the best any unexplored region could still deliver:

$$\overline{z} \;=\; \max\Big\{\, \underline{z},\; \max_{k \text{ open}} z_{\mathrm{LPR}}(k) \,\Big\} \qquad\qquad \underline{z} \;=\; c^\top \hat{x} \;\text{ (incumbent, the lower bound (하한))}$$

but an open node has not been solved — there is no $z_{\mathrm{LPR}}(k)$ yet. use its *parent's* value:

$$P_k \subseteq P_{\mathrm{parent}(k)} \qquad\Longrightarrow\qquad z_{\mathrm{LPR}}(k) \;\le\; z_{\mathrm{LPR}}(\mathrm{parent}(k))$$

fixing one more variable can only shrink the region, so the parent's value is a valid — slightly loose — stand-in

this is exactly what a solver prints as its `Best Bd` column: a bound over *pending* nodes, not a solved LP

the gap, and the stop:

$$\text{gap} \;=\; \frac{\overline{z} - \underline{z}}{|\underline{z}|} \qquad\qquad \text{open list empty} \;\Rightarrow\; \overline{z} = \underline{z} \;\Rightarrow\; \text{gap } 0 \;\Rightarrow\; \text{proved optimal}$$

$\underline{z}$ only rises (better plans found), $\overline{z}$ only falls (promises expire) — the two curves close on z_{IP}

KEY the upper bound is not one LP. it is the *maximum over the open nodes* — and it drops only when the node holding it is finally solved.

3. Minimizing? flip the sign once, at the start

the LAP minimizes cost; week 7 maximizes. the demo flips once — $z = -\,\text{cost}$, $\max z = -\min \text{cost}$ — so every rule above reads unchanged:

	min cost (the model you wrote)	**max** $z = -$**cost** (what the demo solves)
relaxation LP_k	*lower* bound on cost	*upper* bound $\overline{z}$
incumbent $\hat{x}$	*upper* bound on cost	*lower* bound $\underline{z}$

the relaxation is always the **optimistic** side (it may cheat), the incumbent always the **achievable** side (someone can execute it)

so on screen the blue curve falls and the magenta one rises; on your own min-cost model, read them as a cost floor rising and a cost ceiling falling

watch nothing here is Lagrangian duality. a Lagrangian bound would move the hard rows *into* the objective with multipliers λ and optimize what is left; this is the LP relaxation of week 7, §1 — same principle (relax, then bound), different relaxation. a Lagrangian bound on $r_i \le B x_i$ is a fine project topic, and it is never worse than the LP bound.

4. Why the LAP's bound is so weak

the linking row is $r_i \le B\, x_i$ with $M = B$ (week 6). in the relaxation, set

$$x_i \;=\; \frac{r_i}{B} \;\in\; [0,1]$$

and the reserve is fully unlocked for a *fraction* of the opening cost. the LP opens slivers of every site, the fixed costs nearly vanish, and $\overline{z}$ is far too optimistic

week 5's two-site toy: $z_{\text{LP}} = 62$ against $z_{\text{IP}} = 65$, at $x = (\frac{8}{15}, \frac{1}{3})$

at $n = 30$ sites the demo needs roughly 9,000 LPs to close that gap; $n = 50$ runs for minutes

two ways to close a gap, and only one of them is the solver's:

from below	branching finds better plans	$\underline{z}$ rises — search
from above	a tighter P_0 (week 7, §2)	$\overline{z}$ falls — *modeling*

that is the whole argument for alternative formulations: you are not buying a better answer, you are buying a better *bound*, and the bound is what the search spends its time on

demo `branch_and_bound_lap.py` — the bounds panel is this note: magenta = $\underline{z}$ (incumbent), blue = $\overline{z}$ (best open-node bound), one point per LP solved. *node order* switches the rule that picks the next open node — depth-first dives (incumbents early, bound falls late), best-bound takes the node holding $\overline{z}$ (bound falls early, incumbents late). the gap target stops the run early, the way a solver's `MIPGap` does.

takeaway **upper bound = LP relaxation, one per node, maximized over the nodes still open. branching raises the floor; a tighter formulation lowers the ceiling.**